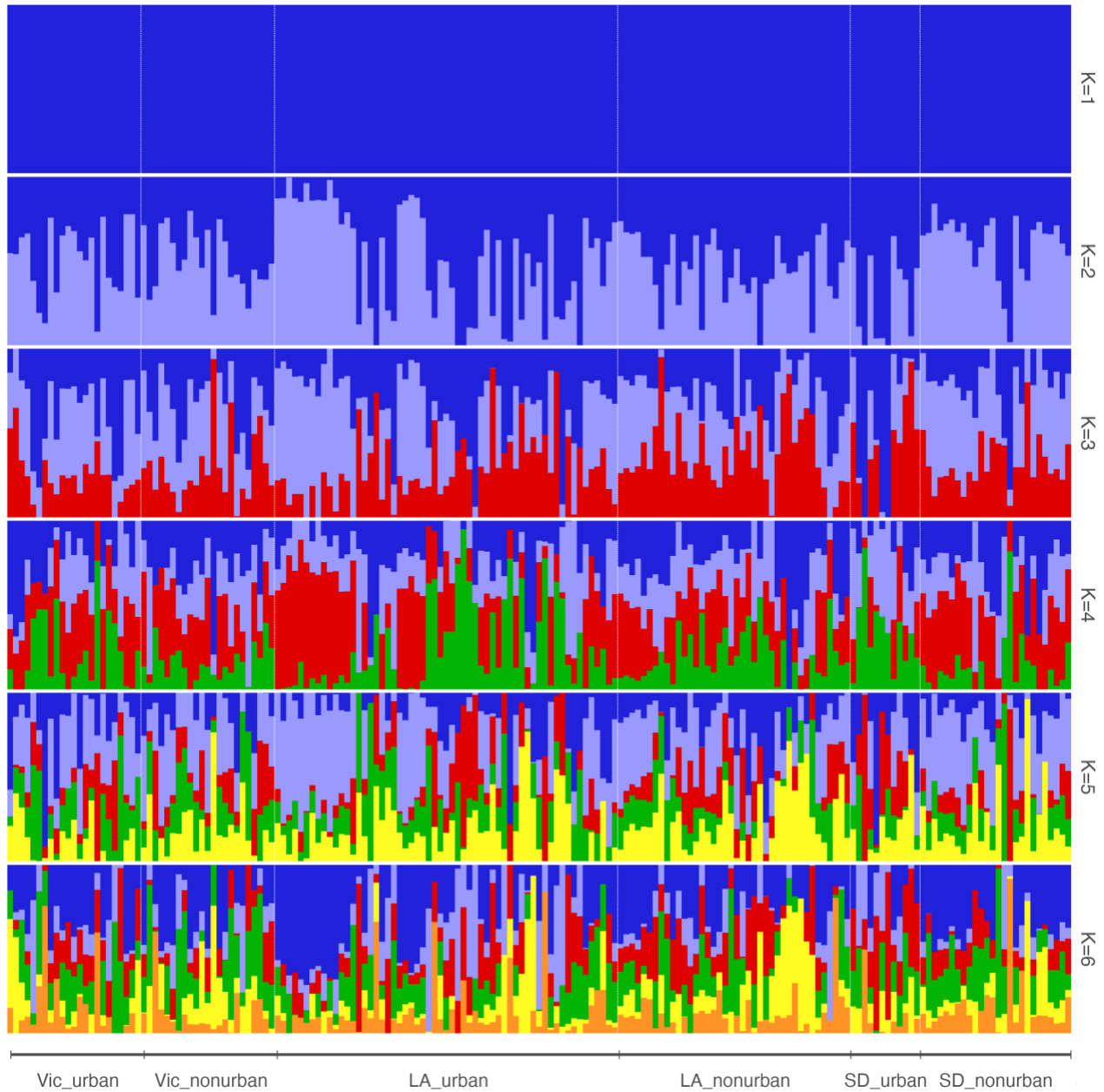


Supplemental Figures:

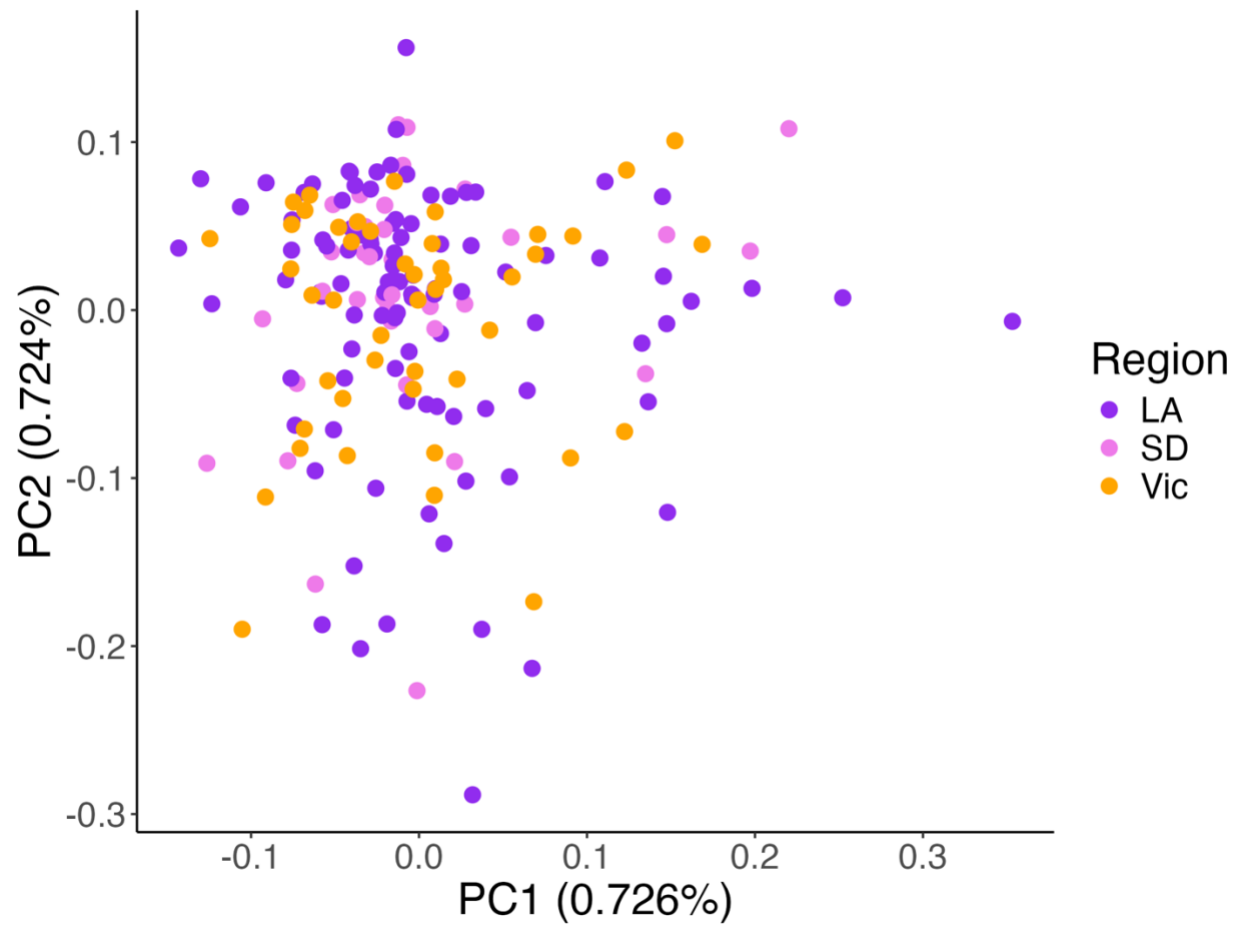
Supp Table 1: Table of sample sites. This includes exact site latitude/longitude and what region they correspond to: Los Angeles (LA), San Diego (SD) or Victoria, B.C. (Vic). The table also includes site ID, tidal zone (intertidal or subtidal), whether the site was a marine protected area (MPA) and who collected the samples by last name.

Site	Latitude	Longitude	region	Urban	Site_ID	Tidal Zone	California MPAs	Collector
Sugarloaf Point, Orange County	33.5197	-117.76448	LA	nonurban	Sug	intertidal	Laguna Beach SMR	Armstrong
Treasure Island Park, Orange County	33.51372	-117.75829	LA	nonurban	Trp	intertidal	Laguna Beach SMR	Armstrong /Sutherland
Twin Points, Laguna Beach	33.54529	-117.80068	LA	nonurban	Twp	intertidal	Laguna Beach SMR	Armstrong /Lee
Laguna Beach	33.52987	-117.78093	LA	nonurban	Lag	subtidal	Laguna Beach SMR	Williams
Recreation Point	33.54119	-117.79276	LA	nonurban	Rec	subtidal	Laguna Beach SMR	Toy
Cabrillo Beach, San Pedro	33.70716	-118.28562	LA	urban	Cab	intertidal	N/A	Armstrong /Kawahara/Dellinger
Portuguese point	33.73837	-118.37577	LA	urban	Port	intertidal	Abalone Cove SMCA	Armstrong /Lee
Wilders Addition Park, San Pedro	33.7096	-118.29956	LA	urban	WAP	intertidal	N/A	Armstrong /Kawahara/Dellinger
White Point, San Pedro	33.71497	-118.31871	LA	urban	Wh	intertidal	N/A	Armstrong /Dellinger
KOU Rock	33.7203	-118.33722	LA	urban	KOU	subtidal	N/A	Scholtz
Pelican Cove	33.74090 35	-118.40399	LA	urban	Pel	subtidal	Abalone Cove SMCA	Wall
Big Rock Reef, San Diego	32.8223	-117.28095	SD	nonurban	Brr	intertidal	South La Jolla SMR	Armstrong

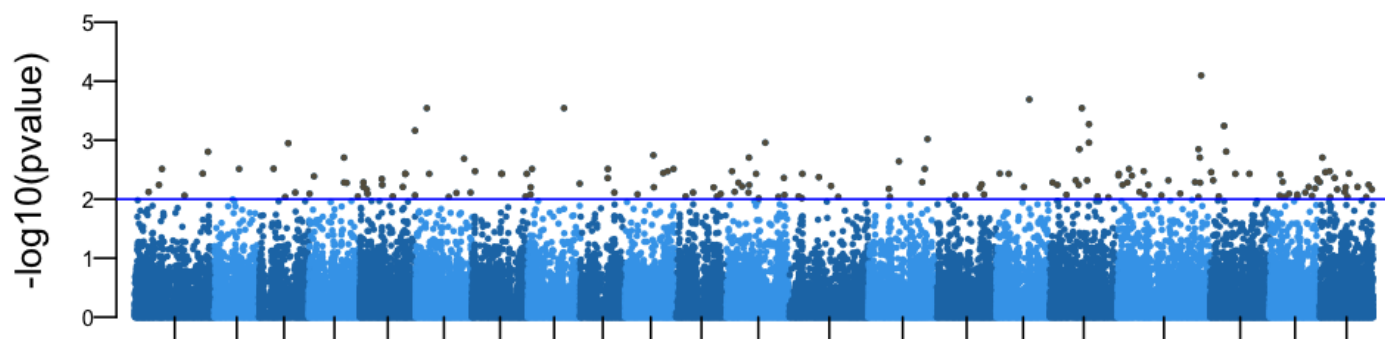
La Jolla, San Diego	32.8196	-117.2871	SD	nonurban	Bir	subtidal	South La Jolla SMR	Parnell
Leucadia	33.06373	-117.31094	SD	nonurban	Leu	subtidal	N/A	Scholtz
Point Loma, San Diego	32.7007	-117.2638	SD	urban	Lom	subtidal	Cabrillo SMR	Parnell
Botanical Beach	48.52918	-124.4525	Vic	nonurban	Bot	intertidal	N/A	Armstrong
Whiffin Spit, Sooke BC	48.35431	-123.72819	Vic	nonurban	Whif	intertidal	N/A	Armstrong
Clover Point, Victoria BC	48.40281	-123.35042	Vic	urban	Clov	intertidal	N/A	Armstrong
Macaulay Point, Victoria BC	48.41659	-123.40953	Vic	urban	Mac	intertidal	N/A	Armstrong



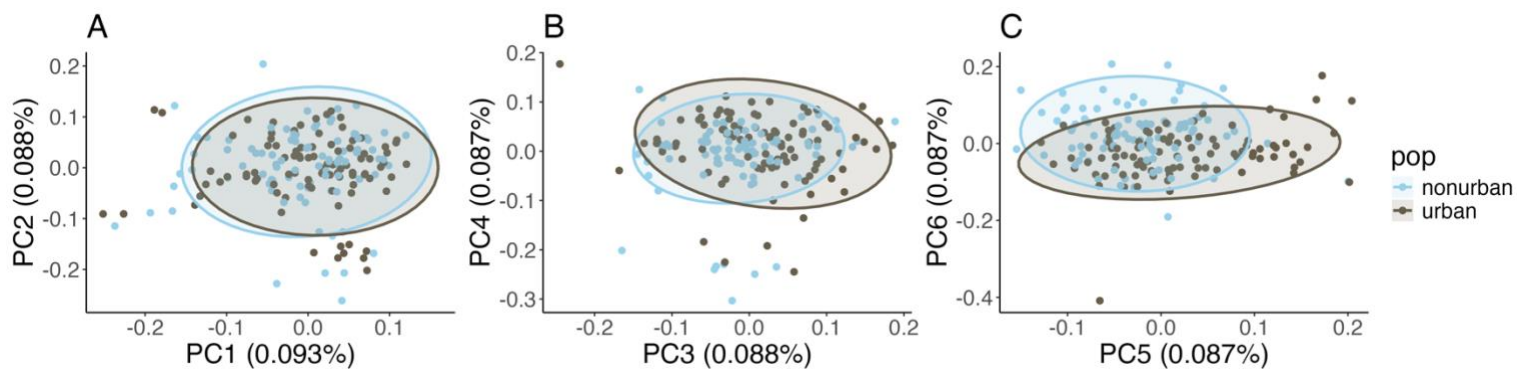
Supp Fig 1. Structure plots for K=1 through K=6 with hierarchical clustering implemented in Tess3r (v. 1.1.0) and Pophelper (v. 2.3.1) for visualization. The samples are organized by region (Victoria, Los Angeles and San Diego), and within each region the samples are separated by urban and nonurban sites. From top to bottom you can see increasing K values. The K that was the best fit for our data was K=1 (top panel).



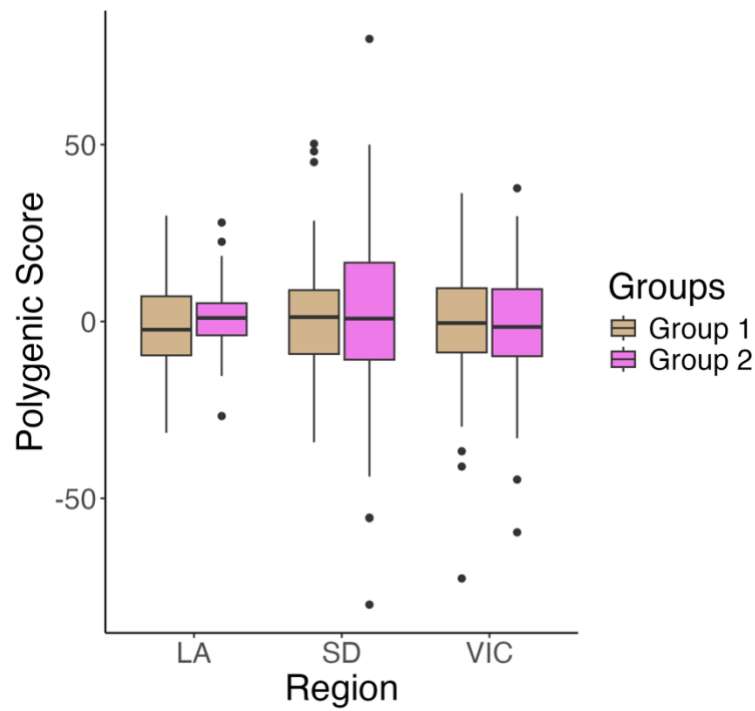
Supp Fig 2. Thinned PCA colored by region: Los Angeles (LA), San Diego (SD) and Victoria (Vic). The axes correspond to principle components capturing 0.726% of the variation in the data (PC1) and 0.724% of the variation in the data (PC2). There is no clear grouping by region in this PCA.



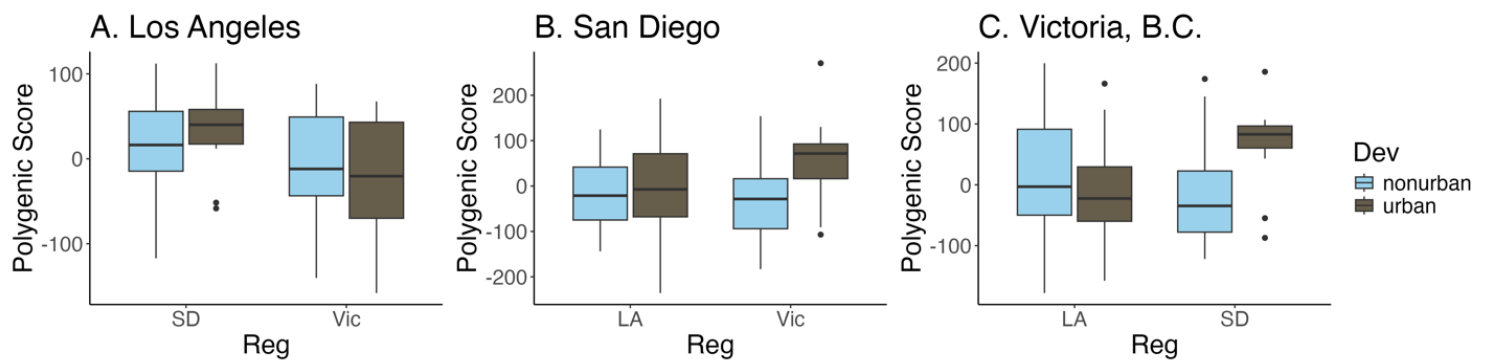
Supp Fig 3. Manhattan Plot using OUTFlank (v. 0.2) for urbanization outliers. The blue line indicates an outlier cutoff of $p < 0.01$, with 165 outliers associated with urbanization.



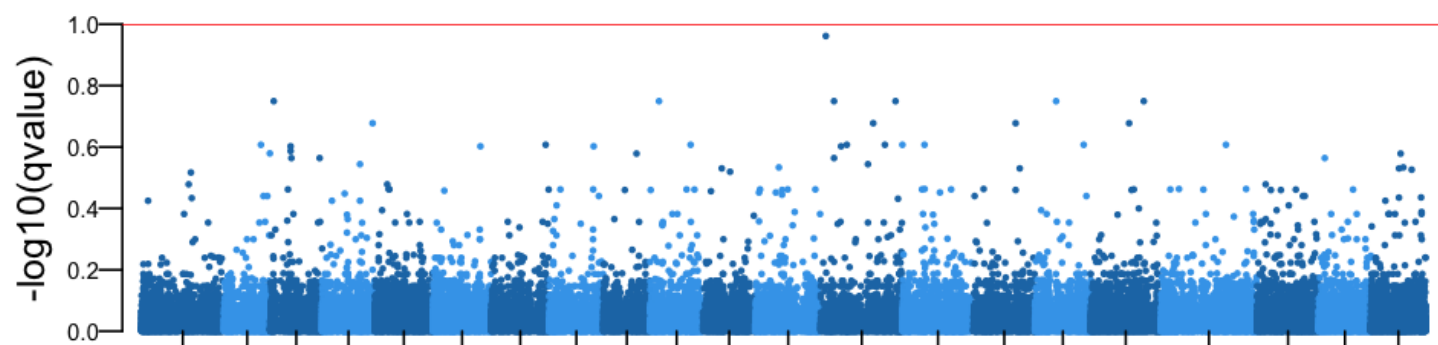
Supp Fig 4. Visualization of PC axes for urban and nonurban samples, with PCs identified using *pcadapt* (v. 4.4.0). A) Urban and nonurban samples significantly differentiated along PC3 (t-value=2.941, p=0.0037). Colors distinguish urban (brown) and nonurban (light blue) samples. B) This differentiation between urban and nonurban samples was observed in all three regions: Los Angeles (LA), San Diego (SD) and slightly in Victoria (Vic).



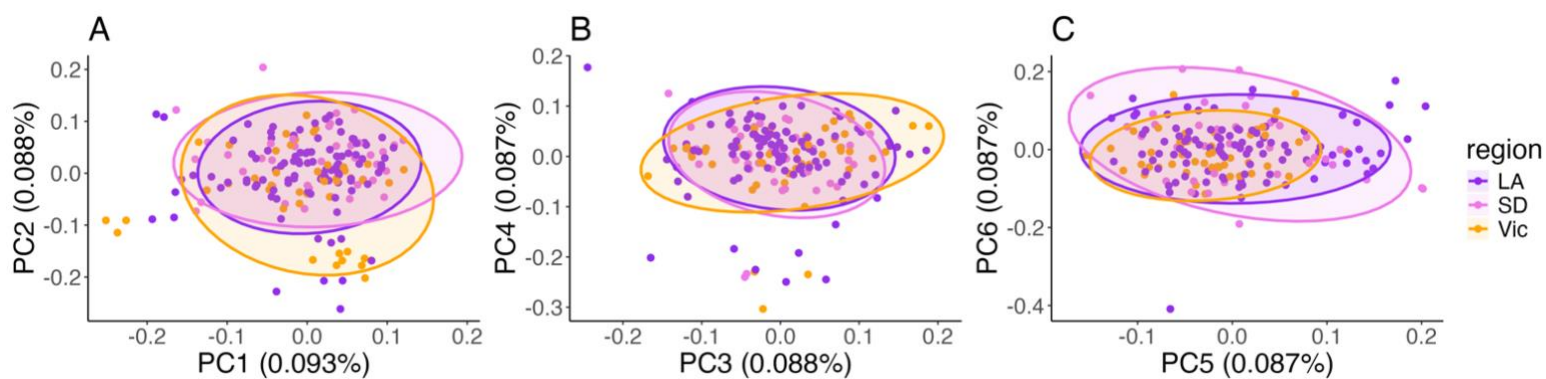
Supp Fig 5. Null model with the labels for urban and nonurban sites randomized for the polygenic score test to assess if the groups were separating by random chance. This null model showed no ability for the polygenic score to differentiate samples, with group 1 (brown) and group 2 (pink), as randomly assigned groups. After 100 runs, there is no significant differentiation from either group, supporting that the separation of our data into two groups is not random chance.



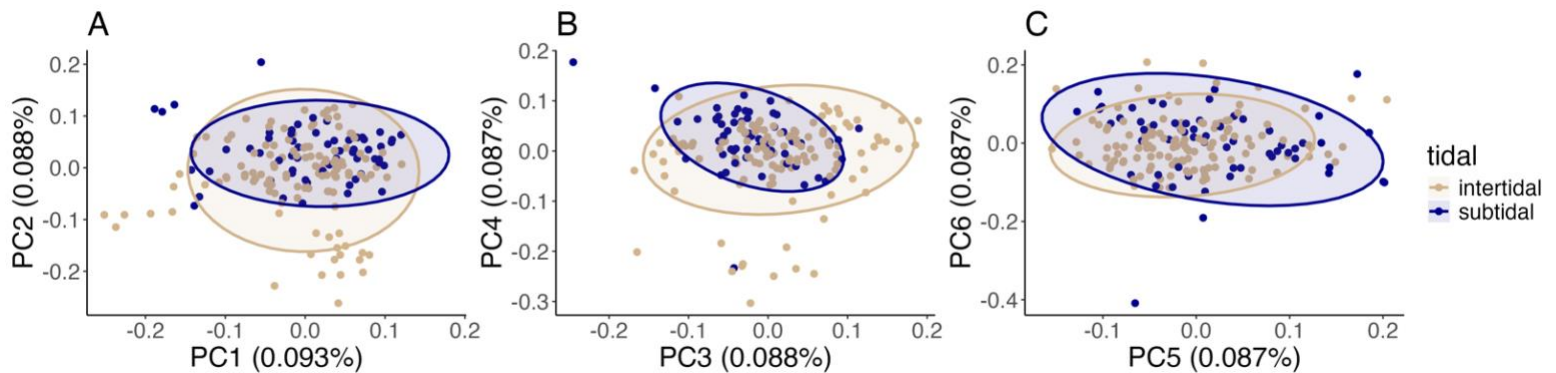
Supp Fig 6. To test the predictive power of one region versus the others for detecting differences between urban/nonurban samples, region and the interaction, we compared one region to the remaining two for all three comparisons. The training region is listed at the top of each boxplot for A-C and the compared regions are boxplots and labeled on the x-axis. The three regions are San Diego (SD), Victoria, B.C. (Vic) and Los Angeles (LA), and colors distinguish urban (brown) and nonurban (light blue) samples. (A) When comparing Los Angeles versus San Diego and Victoria, urban (two-way ANOVA, $F_{1,1} = 0.341$, $p = 0.561$) and the interaction of urban and region (two-way ANOVA, $F_{1,1} = 1.465$, $p = 0.230$) were not significant, while region was significant (two-way ANOVA, $F_{1,1} = 5.721$, $p = 0.019$). (B) When comparing San Diego versus Los Angeles and Victoria, urban (two-way ANOVA, $F_{1,1} = 6.845$, $p = 0.010$) and the interaction of urban and region (two-way ANOVA, $F_{1,1} = 5.493$, $p = 0.020$) were significant. However region was not (two-way ANOVA, $F_{1,1} = 1.976$, $p = 0.162$) (C) Lastly, when comparing Victoria versus Los Angeles and San Diego, neither region (two-way ANOVA, $F_{1,1} = 0.750$, $p = 0.388$) nor urban (two-way ANOVA, $F_{1,1} = 0.079$, $p = 0.779$) were significant, but the interaction of the two was (two-way ANOVA, $F_{1,1} = 9.699$, $p = 0.002$).



Supp Fig 7. Manhattan Plot for OUTFlank (v. 0.2) across regions. The red line indicates that even with an outlier cutoff of $p < 0.1$ there are no outliers that correspond with region for this dataset.



Supp Fig 8. Visualization of PC axes region samples, with PCs identified using *pcadapt* (v. 4.4.0). There was no variation across regions on any of the PC axes tested (A-C). Each region is a distinct color: Los Angeles (LA) in purple, San Diego (SD) in pink and Victoria, B.C. (Vic) in orange.



Supp Fig 9. Visualization of PC axes for subtidal and intertidal samples, with PCs identified using *pcadapt* (v. 4.4.0). Variation at several PC axes observed between samples across different tidal zones. We found that tidal zone drove variation along several PC axes, included PC2 (A, t-value= 2.293, p=0.023) and PC3 (B, t-value=-2.908, p=0.0041) and PC4 (B, t-value=3.588, p<0.001). Colors distinguish intertidal (tan) and subtidal (navy blue) samples. This was not unexpected given previous literature highlighting SNP outliers between subtidal and intertidal *S. purpuratus* populations across California (20).

Supplemental Tables 3-5 of Gene Ontology outputs for each region are stored in the excel spreadsheet GOTerms_full.xlsx